

Real-Time Physics-Based Force Feedback for VR Simulation in Medicine

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ABSTRACT

Tactile feedback in Virtual Reality (VR) is crucial for immersion, yet many systems rely on limited static feedback, reducing training effectiveness, especially in medical simulations. This paper introduces a VR system that utilizes dynamic force feedback through real-time physics calculations. A comparative study showed that while basic models provide predictable feedback, the physics simulation method offers real-time internal tissue behavior and complex force feedback. This approach significantly enhances the user experience and is essential for effective training in high-realism VR fields, particularly in medical environments.

Index Terms: Virtual Reality, Force Feedback, FEM Simulation

1 INTRODUCTION

Medical education, particularly surgical training, entails high costs and potential ethical issues. To overcome these limitations, surgical simulators have emerged as educational tools that enable safe, repeatable, and cost-effective training [4]. While early simulators relied on 2D screens, they have since evolved into immersive environments using Virtual Reality (VR) technology [7]. A comparative study by Grantcharov et al. [2], for example, reported that a group trained in VR performed a laparoscopic cholecystectomy faster and with fewer errors than a control group, while also significantly reducing unnecessary movements. Furthermore, simulation-based training can also enhance non-technical competencies such as communication and decision-making during surgical procedures [8].

Furthermore, VR surgical training simulators that utilize force feedback, even if it is static force, show great promise for developing the tactile memory essential for surgeons. Training that allows surgeons to directly 'feel' tissue resistance or needle penetration maximizes learning effectiveness far beyond visual-only feedback [5]. However, most current medical VR simulators that offer tactile sensations provide feedback based on static or pre-defined parameters. This imposes a significant limitation: the system requires developers to anticipate every possible interaction and manually define the appropriate haptic response for each. Consequently, if a user attempts an unexpected interaction, the simulator gives unrealistic feedback. This unrealistic virtual tactile sensation can be counterproductive to training efficiency, and the resulting decrease in immersiveness can reduce the overall effectiveness of virtual training [5].

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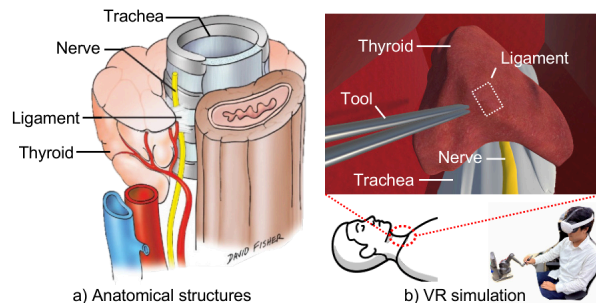


Figure 1: a) An anatomical structure near the neck that includes the trachea, nerve, ligament, and thyroid (Image from [3]). b) VR simulation for validation. A user pulls the thyroid using the tool (tweezers - Touch X).

Therefore, overcoming this limitation requires technology that can simulate the physical behaviors and properties of biological tissue in real-time and convey them as tactile feedback. The Finite Element Method (FEM), a numerical method capable of flexibly computing the complex deformations and precise interactions of biological tissues, presents a possible solution to this issue [6]. In this paper, we present how dynamic physics simulation can create a more realistic and rich force feedback experience compared to conventional static force methods. We demonstrate a comparison result from a system that provides a distinctly different and more varied range of tactile feedback by integrating Unity for the VR environment, the FEM-based SOFA (Simulation Open Framework Architecture) [1] for soft body physics simulation, and a 3D Systems Touch X device for sophisticated 3D force feedback.

2 METHOD

To validate our system, we simulated a pulling situation during a thyroidectomy [3]. During a thyroidectomy, pulling the thyroid also stretches the ligament connecting it to the trachea and the nerve tissue within the ligament (Fig 1a). This stretch can easily cause excessive strain on the nerve, leading to permanent damage, but it is hard to observe directly before pulling. Therefore, our system simulates pulling the thyroid, which is connected by soft tissue (a ligament containing a nerve) to hard tissue (the trachea) during thyroidectomy. This situation is simulated in the Unity-based VR environment, using the SOFA plugin for Unity by InfiniTech3D to perform real-time physics simulations and the Touch X plugin by 3D Systems to deliver force feedback to a haptic device.

We compared force feedbacks generated from these three different modeling conditions under simulation (Fig 2a).

- Rigid Body: All tissues (trachea, ligament, and thyroid) were modeled as non-deformable rigid bodies.

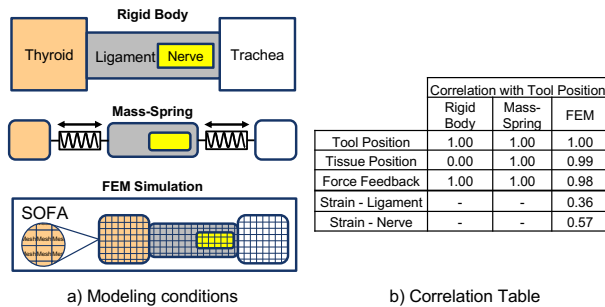


Figure 2: a) Diagrams for each modeling condition. b) Correlation Table with each modeling condition.

- Mass-Spring System: Each tissue was modeled as a simplified mass-spring system.
- FEM Simulation: The physical properties of each tissue and its shape were simulated in real-time using FEM.

Within the same VR environment, the thyroid is pulled in a set direction (perpendicular to the nerve) using a virtual tool operated by the Touch X. To compare each modeling condition, we investigate the relationship between the tool's position and two key variables: the displacement of the tissue objects and the strength of the force feedback. For the FEM simulation, we additionally analyze how the internal strain on the ligament and nerve changed.

3 RESULT

The simulation results for each physical model showed distinct differences in the tissue's response to the virtual tool's position change, the force feedback, and the internal strain (Fig 2b).

In the Rigid Body model, the tissue's position did not react to the tool's movement, showing a correlation of 0. In contrast, the force feedback had a perfect linear relationship (correlation = 1) with the tool's displacement. This implies an interaction where resistance changed predictably, resembling pulling a fixed object to the user.

For the Mass-Spring model, the tissue's position and the force feedback showed a perfect linear correlation (correlation = 1) with the tool's displacement. Thus, the tool's interaction moved the tissue object in this model, like pulling the tissue with tweezers. However, the force feedback increased in a simple linear fashion, failing to represent the distinct physical properties of the connected tissues, and thus the feedback provided can be unrealistic.

In contrast, the FEM simulation model demonstrated a qualitatively different pattern from the other two models. While the tissue's position and force feedback showed very high correlations with the tool's displacement (0.99 and 0.98, respectively), it is not a perfect linear relationship like the other models. This indicates the presence of nonlinear resistance resulting from the material properties and intricate shape transformations, and that it can generate more realistic feedback than the previous models.

Additionally, the strain results have low correlation between the strain on the ligament and nerve and the tool's position, measured at 0.36 and 0.57, respectively. This implies that, in addition to the direct force applied by the tool, complex interactions with connected tissues significantly impact internal deformation. Thus, the VR simulation can capture realistic internal physical behavior that simpler models cannot predict thanks to the physical simulation.

4 LIMITATION AND FUTURE WORK

While this study validated the quantitative advantages of FEM-based haptics, a user study involving medical professionals was not conducted; instead, it focused on the technical properties. Future

work will involve a comprehensive clinical study with surgeons to evaluate training efficacy and to extend the simulation to include topological changes, such as cutting, with validation against real-tissue data.

5 CONCLUSION

This study shows that choosing the right physics simulation method is crucial for quality haptic feedback. Simple models like Rigid Body or Mass-Spring can give basic tactile cues but have limitations. However, the rigid body model struggles to provide information beyond simple contact, and the mass-spring model is unable to reproduce physically plausible deformations, showing limitations for realistic training. In contrast, the simulation based on the FEM was able to provide realistic force feedback that included complex tissue deformation and internal resistance. Particularly in medical training, which requires delicate interaction with multiple soft tissues, the integration of physics simulations like FEM is essential for providing advanced force feedback. This research helps VR developers understand the significant differences between simplified methods and real physics to improve feedback.

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